

# Momentum, Reversal, Volatility and Size in U.S. Large Caps

A cross-sectional factor study with Fama-MacBeth validation, 2011 to 2026: one-page summary

Zack Higham, BSc Mathematics, Queen Mary University of London. Full paper (23 pages) and code:

[github.com/zack-higham/Cross-Sectional-Factor-Research-with-Fama-MacBeth-Validation](https://github.com/zack-higham/Cross-Sectional-Factor-Research-with-Fama-MacBeth-Validation)

## Question

Do four standard characteristics (12-1 momentum, one-month reversal, 60-day total volatility, log market capitalisation) predict next-month returns among the 200 largest S&P 500 stocks, and which apparent premia survive risk adjustment, costs and checks for sample-construction bias?

## Data and method

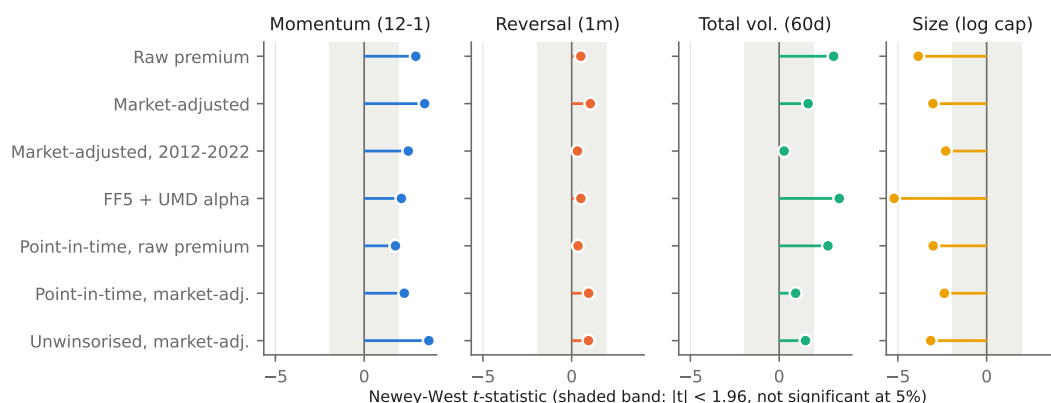
- **Data:** daily adjusted prices from January 2011; 175 months with all four factors, to August 2026; SPY and T-bills; Fama-French five factors and UMD; historical S&P 500 membership.
- **Three lenses:** equal-weighted decile long-short portfolios, Fama-MacBeth regressions with Newey-West  $t$ -statistics, and rank information coefficients, all on one timing convention (factor at end of  $t - 1$ , return over  $t$ ).
- **Validation:** out-of-sample split, CAPM and FF5+UMD alphas, 10 bps costs, drawdowns, sector neutrality, VIX regimes, point-in-time membership and unwinsorised factors (the last four committed before being run).

## Findings

1. **Momentum: the only positive premium significant after market adjustment and point-in-time.** Market-adjusted Fama-MacBeth premium 3.7% a year per standard deviation ( $t = 3.41$ ;  $t = 2.49$  before 2023). Under point-in-time membership the market-adjusted premium stays significant ( $t = 2.26$ ), but the unadjusted premium ( $t = 1.76$ ) and the decile spread ( $t = 0.41$ ) do not. It loads heavily on UMD ( $t = 8.83$ ), and its decile portfolio's Carhart alpha is  $-1.48\%$  a year, so it is not a separate source of return. The decile portfolio lost 56.3% from January 2016 to May 2023.
2. **Reversal is eliminated by costs.** Its raw premium is insignificant (Fama-MacBeth  $t = 0.51$ ), and its turnover (3.44 a month) puts its break-even cost at 9.7 bps, below realistic trading costs.

## Diagnosed artefacts of the sample

3. **Volatility: market beta under the CAPM, selection under FF5+UMD.** Raw Fama-MacBeth  $t = 3.07$ , but the long-short portfolio has a market beta of 1.17 and an insignificant CAPM alpha ( $-3.13\%$  a year before 2023). Its FF5+UMD alpha of 15.95% a year ( $t = 3.51$ ) comes from unprofitable, heavily investing firms that nonetheless won: the signature of ex post selection. The residual volatility premium loses significance under point-in-time membership ( $t = 3.05$  to 1.79).
4. **Size: survivorship bias, not a size effect.** The big-minus-small spread ( $-21.26\%$  a year) comes from an end-of-sample universe; the point-in-time filter shrinks it by 25.4%, a partial confirmation under the pre-set rule.



Fama-MacBeth  $t$ -statistics across specifications. Momentum is significant in all but the unadjusted point-in-time premium ( $t = 1.76$ ); size is significantly negative throughout (survivorship bias).

## Main limitation

The universe is the September 2026 top 200, applied to the whole history. The point-in-time filter removes stocks from months before they joined the index, but firms that left it (acquired, shrank, failed) are still missing, so the remaining survivorship bias is smaller but not zero. Results rest on 175 months and about 20 stocks per decile, and no premium outside 2023 to 2026 clears the  $t > 3$  multiple-testing hurdle.